AWS Secure Infrastructure with OpenVPN, S3, CloudFront and WAF

Project Report

1. Abstract

This project demonstrates a secure AWS infrastructure using a Virtual Private Cloud (VPC), Public and WAF

2. Objectives

- Build a secure cloud environment.
- Implement private network access using OpenVPN.
- Host a static website on Amazon S3.• Deliver content globally using CloudFront.
- Protect applications using AWS WAF.
- Implement IAM-based access control.

3.**Architecture** Overview

```
Internet Users
      |
  CloudFront
      |
      WAF
      |
  S3 Website

OpenVPN Server
      |
  VPN Tunnel
      |
  Private EC2

VPC → Subnets → Route Tables → Security Groups → IAM
```

4. AWS Services Used

VPC, Subnets, Internet Gateway,
Route Tables, Security Groups,
EC2, OpenVPN Access Server, IAM Users, IAM Roles,
Amazon S3, CloudFront, Origin Access Control (OAC),
AWS WAF, CloudWatch, Systems Manager.

5. Implementation

Phase 1: Networking Setup

- Create VPC (10.0.0.0/16)
- Create Public and Private Subnets
- Attach Internet Gateway
- Configure Route Tables

Phase 2: OpenVPN & Private EC2

- Deploy OpenVPN in Public Subnet
- Deploy Private EC2 without Public IP
- Access EC2 through VPN only

Phase 3: Static Website Hosting

- Create S3 Bucket
- Upload Website Files
- Enable Static Website Hosting

Phase 4: CloudFront CDN

- Create Distribution
- Configure OAC
- Test Global Access

Phase 5: AWS WAF

- Create Web ACL
- Add Managed Rules
- Enable SQL Injection and XSS Protection
- Configure Rate Limiting

6. Security Features

- Network Isolation using VPC
- Private EC2 without Public Access
- VPN-based Administrative Access
- IAM Role-Based Permissions
- S3 Secure Access using OAC
- AWS WAF Protection
- CloudWatch Monitoring

7. Testing & Validation

- VPN Connection Test
- SSH Access through VPN
- Website Accessibility Test
- CloudFront Distribution Test
- SQL Injection Protection Test
- XSS Protection Test
- Rate Limiting Validation

8. Results

The infrastructure successfully provides:

- Secure administrative access
- Private workload protection
- Highly available website hosting
- Global content delivery
- Web application security
- IAM-based access management

9. Conclusion

This project successfully implements a secure AWS cloud infrastructure using VPC, OpenVPC,EC2,S3,CloudFront,AWS WAF and IAM. It provides secure remote access ,reliable website hosting ,global content delivery and protection against common web attacks. The project demonstrates AWS security best practices and real-world cloud deployment techniques.

10. References

- AWS Documentation
- OpenVPN Documentation
- Amazon S3 Documentation
- CloudFront Documentation
- AWS WAF Documentation
- AWS IAM Documentation